Sample Paper 04

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

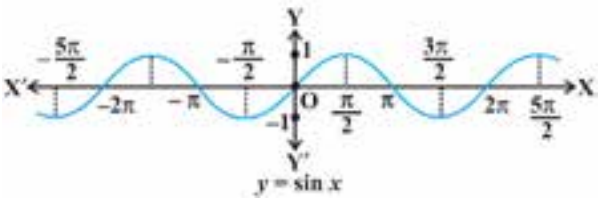
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the period of the sine curve as displayed in the image?



(a) π (b) 2π

(c) 3π (d) 4π

2. The solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is

(a) $e^x + e^{-y} + k = 0$ (b) $e^{2x} = ke^y$

(c) $e^x = ke^{2y}$ (d) $e^x = ke^y$

3. In the set of all straight lines in a plane, the relation R "to be Perpendicular" is-

(a) Reflexive and transitive

(b) Symmetric and transitive

(c) Symmetric

(d) None of these

4. The modulus of the vector $19\hat{i} + 5\hat{j} - 6\hat{k}$ is.

(a) $\sqrt{322}$ (b) $\sqrt{420}$

(c) $\sqrt{421}$ (d) $\sqrt{422}$

5. Required distance $= \sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{41}$

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, then which one of the following is equal to A' ?

(a) $\begin{bmatrix} 2 & 1 & 3 \\ 5 & 4 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ (d) $\begin{bmatrix} 3 & 6 \\ 2 & 5 \\ 1 & 4 \end{bmatrix}$

6. $\frac{d(2^x)}{d(3^x)} =$

(a) $\left(\frac{2}{3}\right)^x$ (b) $\frac{2^{x-1}}{3^{x-1}}$

(c) $\left(\frac{2}{3}\right)^x \log_3 2$ (d) $\left(\frac{2}{3}\right)^x \log_2 3$

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7. $\int \frac{dx}{x+\sqrt{x}}$

(a) $\log x + \log(1+\sqrt{x}) + C$

(b) $2\log(1+\sqrt{x}) + C$

(c) $\log(1+\sqrt{x}) + C$

(d) $\log\sqrt{x} + C$

8. If a particle moves in a straight line so that $s = \frac{1}{2}vt$, then acceleration is

(a) a constant (b) proportional to t

(c) proportional to v (d) proportional to s

9. The maximum value of $f(x) = \frac{\log x}{x}$ is.

(a) 1 (b) $\frac{2}{e}$

(c) e (d) $\frac{1}{e}$

10. Evaluate $\begin{bmatrix} 1 & -1 \\ y & x \end{bmatrix}$

(a) $x+y$ (b) $x-y$

(c) $-y-x$ (d) $1-x$

11. $\frac{d}{dx}[\sin^{-1}x + \cos^{-1}x] = ?$

(a) 0 (b) $\frac{1}{\sqrt{1-x^2}}$

(c) $-\frac{1}{\sqrt{1-x^2}}$ (d) $\frac{1}{2}\sqrt{1-x^2}$

12. Which one of the following statements is correct

(a) Every L.P.P admits an optimal solution

(b) A L.P.P admits unique optimal solution

(c) The optimal value occurs at a corner point of the feasible region

(d) If a L.P.P admits two optimal solutions, then it has infinite optimal solutions

13. The distance of the point $(3,4,5)$ from x-axis is-

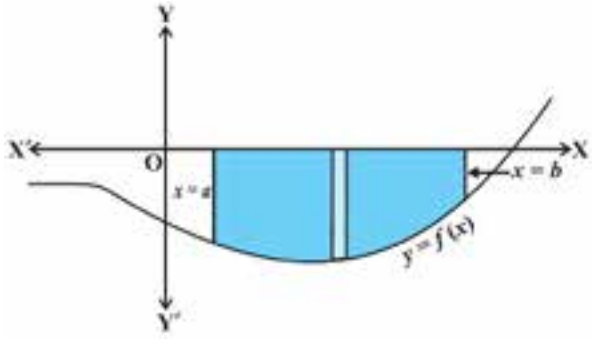
(a) 3

(b) 5

(c) $\sqrt{41}$

(d) None of these

14. In Fig., which statement correctly describes the position of the curve y=f (x)?



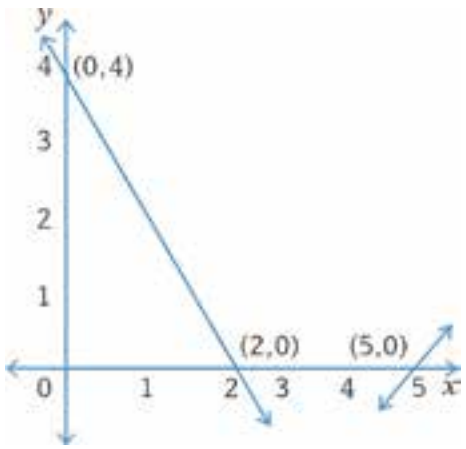
(a) It lies entirely above the X-axis.

(b) It lies entirely below the X-axis.

(c) It intersects the Y-axis only at one point.

(d) It passes through both X and Y axes at multiple points.

15. For the LPP shown in the figure, the objective is Minimise $Z = 3x + 2y$. From the graph, the minimum value of Z is obtained at the corner point (2,0). The optimal solution of this LPP:



(a) Does not exist as the feasible region is unbounded.

(b) Does not exist as the inequality 3x+2y<6 does not have any point in common with the feasible region.

(c) Exists as the inequality 3x+2y>6 has infinitely many points in common with the feasible region.

(d) Exists as the inequality 3x+2y < 6 does not have any point in common with the feasible region.

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16. $\sin^{-1}\frac{1}{x} = ?$

(a) $\sec^{-1}x$ (b) $\operatorname{cosec}^{-1}x$

(c) $\tan^{-1}x$ (d) $\sin x$

17. $\int \frac{dx}{1+x^2} = ?$

(a) $\tan x + c$ (b) $\tan^{-1}x + c$

(c) $\cot x + c$ (d) $-\cot^{-1}x + c$

18. The coordinates of the midpoint of the line segment joining the points (2, 3, 4) and $(8, -3, 8)$ are

(a) (10, 0, 12) (b) (5, 6, 0)

(c) (6, 5, 0) (d) (5, 0, 6)

19. **Assertion :** If A and B be two events corresponding to sample space S such that $P(A) = 0.2$ and $P(B) = 0.8$, then $A \cup B$ is a sure event.

Reason : If A and B are mutually exclusive events, then $P(A) + P(B) = 1$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

20. **Assertion :** $f(x) = x^3$ is continuous at $x = 2$.

Reason: $f(x)$ is continuous at $x = a$ if

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Differentiate $\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$ with respect to x.

OR

If $y = \sin^{-1}(6x\sqrt{1-9x^2})$, $-\frac{1}{3\sqrt{2}} < x < \frac{1}{3\sqrt{2}}$, then find $\frac{dy}{dx}$.

22. Suppose a girls throws a die. If she gets 1 or 2, she tosses a coin three times and notes the number of tails. If she gets 3, 4, 5 or 6, she tosses a coin once gets notes whether a 'head' or 'tail' is obtained. If she obtained exactly one 'tail', what is the probability that she threw 3, 4 ,5 or 6 with the die?

23. Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60° and their scalar product is $\frac{9}{2}$.

24. Find the general solution of differential equation $y = e^{2x}(a + bx)$

25. Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$, then find $(A')^{-1}$.

OR

Find the adjoint of the matrix $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

and hence show that $A(adj\ A) = |A| I_3$.

27. Find the values of a and b such that the function defined as follows is continuous.

$$f(x) = \begin{cases} x+2, & x \le 2 \\ ax+b, & 2 < x < 5 \\ 3x-2, & x > 5 \end{cases}$$

28. Let R be a relation defined on the set of natural numbers N as follow:

$$R = \{(x, y) : x \in N, y \in N \text{ and } 2x + y = 24\}$$

Find the domain and range of the relation R. Also, find if R is an equivalence relation or not.

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29. Find a vector of magnitude 5 units and parallel to the resultant of $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$.

30. Find $\int \frac{2x}{(x^2+1)(x^2+2)^2}\,dx$.

OR

Integrate w.r.t. x, $\frac{x^2-3x+1}{\sqrt{1-x^2}}$.

31. Solve the following differential equation

$$x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the value of the following

$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

OR

Find the value of the following $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$

33. Find both the maximum value and minimum value of $3x^4 - 8x^3 + 12x^2 - 48x + 25$ on the interval $[0,3]$.

OR

At what points in the interval $[0,2\pi]$, does the function $\sin 2x$ attain its maximum value?

34. Maximize $Z = -x + 2y$, Subject to the constraints:

$$x \geq 3,\ x + y \geq 5,\ x + 2y \geq 6,\ y \geq 0$$

OR

Minimise $Z = x + 2y$ subject to $2x + y \geq 3$, $x + 2y \geq 6$, x, $y \geq 0$. Show that the minimum of Z occurs at more than two points.

35. Find the value of λ, so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. Also, find whether the lines are intersecting or not.

OR

Find the shortest distance between the lines

$$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$$

and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$.

SECTION - E

Case study based questions are compulsory.

36. Sun Pharmaceutical Industries Limited is an Indian multinational pharmaceutical company headquartered in Mumbai, Maharashtra, that manufactures and sells pharmaceutical formulations and active pharmaceutical ingredients in more than 100 countries across the globe.

Sun Pharmaceutical produces three final chemical products P_1, P_2 and P_3 requiring mixup of three raw material chemicals M_1, M_2 and M_3. The per unit requirement of each product for each material (in litres) is as follows:

$$A = \begin{array}{c} \\ P_1 \\ P_2 \\ P_3 \end{array}\begin{array}{c} \begin{array}{ccc} M_1 & M_2 & M_3 \end{array} \\ \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & 5 \\ 2 & 4 & 2 \end{bmatrix} \end{array}$$



(i) Find the total requirement of each material if the firm produces 100 litres of each product,

(ii) Find the per unit cost of production of each product if the per unit of materials M_1, M_2 and M_3 are ₹ 5, ₹ 10 and ₹ 5 respectively, and

(iii) Find the total cost of production if the firm produces 200 litres of each product.

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37. Quality assurance (QA) testing is the process of ensuring that manufactured product is of the highest possible quality for customers. QA is simply the techniques used to prevent issues with product and to ensure great user experience for customers.



A manufactured component has its quality graded on its performance, appearance, and cost. Each of these three characteristics is graded as either pass or fail. There is a probability of 0.40 that a component passes on both appearance and cost. There is a probability of 0.35 that a component passes on both performance and appearance. There is a probability of 0.31 that a component passes on all three characteristics. There is a probability of 0.64 that a component passes on performance. There is a probability of 0.19 that a component fails on all three characteristics. There is a probability of 0.06 that a component passes on appearance but fails on both performance and cost.

(i) What is the probability that a component passes on cost but fails on both performance and appearance?

(ii) If a component passes on both appearance and cost, what is the probability that it passes on all three characteristics?

(iii) If a component passes on both performance and appearance, what is the probability that it passes on all three characteristics?

38. Commodity prices are primarily determined by the forces of supply and demand in the market. For example, if the supply of oil increases, the price of one barrel decreases. Conversely, if demand for oil increases (which often happens during the summer), the price rises. Gasoline and natural gas fall into the energy commodities category.



The price p (dollars) of each unit of a particular commodity is estimated to be changing at the rate

$$\frac{dp}{dx} = \frac{-135x}{\sqrt{9+x^2}}$$

where x (hundred) units is the consumer demand (the number of units purchased at that price). Suppose 400 units $(x = 4)$ are demanded when the price is \$30 per unit.

(i) Find the demand function $p(x)$.

(ii) At what price will 300 units be demanded? At what price will no units be demanded?

(iii) How many units are demanded at a price of \$20 per unit?

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